NInfer test inventory

Every test in this tree, what it is for, and what runs it. Generated from the files that own each fact: the C++ list from `ctest -N` on `build-test`, the gates from `.githooks/pre-commit`, the triggers from the two workflows, the exceptions from `tools/release/test_baseline.json`, and the probes from `tools/bench` and `tools/scripts`. **174 C++ tests, 39 Python test files.**

What runs when

Trigger	Events	What it does
`ci.yml`	push, pull_request (windows-latest)	runs `bash .githooks/pre-commit` on windows-latest, so the fast gates have one definition
`gpu.yml`	workflow_dispatch ([self-hosted, windows, gpu-5090])	runs the C++ suite and the baseline gate on the self-hosted 5090 runner, serialised against measurements
`.githooks/pre-commit`	every `git commit` here	17 gate scripts, both node plugin checks, pytest, ruff and mypy. The fast tier: seconds, no network.
Manual	on demand	The C++ suite, the sanitizer recipes, and every probe below.

The pre-commit hook, in order

184 steps. A gate that fails stops the commit; `--no-verify` skips it once.

1. `interpreter $PY`
2. `converter tests`
3. `ruff`
4. `mypy`
5. `lane figures`
6. `ok`

> **Gates that deny rather than report:** 17 scripts, each earning its place by catching something a person missed. The retired-names gate, for instance, fails a criterion that names a symbol the tree no longer defines, because such a criterion does not error - it answers a different question.

The C++ suite: 174 tests

Run by `tools/scripts/test_v3.cmd` (which configures, builds and runs), then judged by `tools/release/check_test_baseline.py` against the recorded baseline. Colour marks the exceptions: required, known failure, optional. Failures in any test, required or not, fail the gate.

Test	What it is for
ninfer_public_api_test	The public C++ surface: headers compile and the documented entry points link.
ninfer_cpu_system_test	The CPU execution layer: affinity, worker pools, the executor.

ninfer_cpu_task_pool_test	The CPU execution layer: affinity, worker pools, the executor.
ninfer_cpu_executor_test	The CPU execution layer: affinity, worker pools, the executor.
ninfer_device_test	Device primitives: allocation, synchronisation, the spin path.
ninfer_device_sync_spin_test	Device primitives: allocation, synchronisation, the spin path.
ninfer_device_sync_blocking_test	Device primitives: allocation, synchronisation, the spin path.
ninfer_device_sync_yield_test	Device primitives: allocation, synchronisation, the spin path.
ninfer_device_sync_auto_test	Device primitives: allocation, synchronisation, the spin path.
ninfer_device_sync_invalid_test	Device primitives: allocation, synchronisation, the spin path.
ninfer_device_sync_empty_test	Device primitives: allocation, synchronisation, the spin path.
ninfer_decode_graph_test	An Op, model or runtime test named for what it covers.
ninfer_tensor_test	An Op, model or runtime test named for what it covers.
ninfer_memory_test	An Op, model or runtime test named for what it covers.
ninfer_arena_test	An Op, model or runtime test named for what it covers.
ninfer_kv_cache_test	An Op, model or runtime test named for what it covers.
ninfer_host_context_arena_test	An Op, model or runtime test named for what it covers.
ninfer_state_store_test	An Op, model or runtime test named for what it covers.
ninfer_gdn_replay_records_test	An Op, model or runtime test named for what it covers.
ninfer_host_timing_test	An Op, model or runtime test named for what it covers.
ninfer_uint128_test	An Op, model or runtime test named for what it covers.
ninfer_jinja_test	An Op, model or runtime test named for what it covers.
ninfer_chat_templates_test	An Op, model or runtime test named for what it covers.
ninfer_artifact_reader_test	Artifact framing and binding: containers, codecs, readers, the writer interop.
ninfer_artifact_transfer_plan_test	Artifact framing and binding: containers, codecs, readers, the writer interop.
ninfer_artifact_materialization_test	Artifact framing and binding: containers, codecs, readers, the writer interop.
ninfer_artifact_writer_interop_test	Artifact framing and binding: containers, codecs, readers, the writer interop.
ninfer_scheduler_test	An Op, model or runtime test named for what it covers.
ninfer_context_cost_test	An Op, model or runtime test named for what it covers.
ninfer_resource_manager_test	An Op, model or runtime test named for what it covers.
ninfer_kv_capacity_test	An Op, model or runtime test named for what it covers.
ninfer_sampling_defaults_test	An Op, model or runtime test named for what it covers.
ninfer_grammar_test	An Op, model or runtime test named for what it covers.
ninfer_regex_choice_test	Regex and choice constrained decoding, with its oracle.
ninfer_regex_choice_oracle_test	An Op, model or runtime test named for what it covers.
ninfer_json_schema_test	An Op, model or runtime test named for what it covers.
ninfer_json_schema_oracle_test	JSON-schema constrained decoding, against an oracle.

ninfer_media_decode_test	An Op, model or runtime test named for what it covers.
ninfer_media_acquire_test	An Op, model or runtime test named for what it covers.
ninfer_prompt_input_test	An Op, model or runtime test named for what it covers.
ninfer_pretty_logging_test	An Op, model or runtime test named for what it covers.
ninfer_perplexity_evaluation_test	An Op, model or runtime test named for what it covers.
ninfer_topk_record_test	Top-k scoring records and the log-probability Op.
ninfer_cli_options_test	An Op, model or runtime test named for what it covers.
ninfer_openai_schema_test	An Op, model or runtime test named for what it covers.
ninfer_openai_responses_test	An Op, model or runtime test named for what it covers.
ninfer_openai_responses_store_test	An Op, model or runtime test named for what it covers.
ninfer_anthropic_schema_test	An Op, model or runtime test named for what it covers.
ninfer_serve_options_test	The serving surface: request logs, cancellation, protocol contracts.
ninfer_request_log_test	An Op, model or runtime test named for what it covers.
ninfer_metrics_test	An Op, model or runtime test named for what it covers.
ninfer_http_error_handler_test	An Op, model or runtime test named for what it covers.
ninfer_http_transport_test	An Op, model or runtime test named for what it covers.
ninfer_bench_support_test	Benchmark fixtures and the bench support layer.
ninfer_bench_fixtures_test	Benchmark fixtures and the bench support layer.
ninfer_context_cost_measure_test	An Op, model or runtime test named for what it covers.
ninfer_ngram_graph_planning_test	An Op, model or runtime test named for what it covers.
ninfer_qwen3_5_loading_real_test	OPTIONAL: an operator tool taking --artifact; returns 77 under ctest. **OPTIONAL**
ninfer_qwen3_5_loading_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_frontend_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_runtime_mechanisms_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_state_image_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_state_image_layout_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_context_store_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_native_transactions_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_prefix_real_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_preemption_real_test	The model: binding, frontend, program, state images, speculative rounds.

ninfer_qwen3_5_grammar_real_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_issue5_race_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_tools_real_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_score_real_test	REQUIRED: real causal scoring. **REQUIRED**
ninfer_qwen3_5_vision_workspace_test	REQUIRED: the vision path's workspace. **REQUIRED**
ninfer_qwen3_5_dflash2_real_test	REQUIRED: the real DFlash2 speculative route against the shipped artifact. **REQUIRED**
ninfer_qwen3_5_dflash_prefill_real_test	REQUIRED: real prefill through the DFlash route. **REQUIRED**
ninfer_qwen3_5_moe_real_test	KNOWN FAILURE: needs a 35B-A3B checkpoint; every artifact here is 27B dense. **KNOWN FAILURE**
ninfer_qwen3_5_dflash_real_test	KNOWN FAILURE: needs a `dflash` component; the artifacts ship `dflash2`. **KNOWN FAILURE**
ninfer_tool_call_parser_test	An Op, model or runtime test named for what it covers.
ninfer_qwen3_5_visual_scatter_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_agent_continuation_real_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_tool_constraints_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_qwen3_5_tool_schema_oracle_test	The model: binding, frontend, program, state images, speculative rounds.
ninfer_release_launcher_generation_test	An Op, model or runtime test named for what it covers.
ninfer_add_bias_test	An Op, model or runtime test named for what it covers.
ninfer_gelu_test	An Op, model or runtime test named for what it covers.
ninfer_silu_mul_test	An Op, model or runtime test named for what it covers.
ninfer_residual_add_test	An Op, model or runtime test named for what it covers.
ninfer_sigmoid_mul_test	An Op, model or runtime test named for what it covers.
ninfer_rmsnorm_pack_tail_test	An Op, model or runtime test named for what it covers.
ninfer_l2norm_test	An Op, model or runtime test named for what it covers.
ninfer_kimi_delta_attention_test	An Op, model or runtime test named for what it covers.
ninfer_causal_conv1d_silu_test	Causal attention plans and partitions.
ninfer_layer_norm_test	An Op, model or runtime test named for what it covers.
ninfer_embedding_test	An Op, model or runtime test named for what it covers.
ninfer_argmax_test	An Op, model or runtime test named for what it covers.
ninfer_rope_test	An Op, model or runtime test named for what it covers.
ninfer_embedding_interpolate_add_test	An Op, model or runtime test named for what it covers.
ninfer_sampling_test	An Op, model or runtime test named for what it covers.
ninfer_scalar_test	An Op, model or runtime test named for what it covers.

ninfer_cast_test	An Op, model or runtime test named for what it covers.
ninfer_prepare_ragged_prefix_test	An Op, model or runtime test named for what it covers.
ninfer_scatter_test	An Op, model or runtime test named for what it covers.
ninfer_scatter_bf16_batch_test	An Op, model or runtime test named for what it covers.
ninfer_target_logprobs_test	An Op, model or runtime test named for what it covers.
ninfer_topk_logprobs_test	Top-k scoring records and the log-probability Op.
ninfer_position_test	An Op, model or runtime test named for what it covers.
ninfer_linear_topk_test	An Op, model or runtime test named for what it covers.
ninfer_candidate_selector_test	An Op, model or runtime test named for what it covers.
ninfer_dflash2_nvfp4_routes_test	An Op, model or runtime test named for what it covers.
ninfer_softmax_attention_test	Attention across every KV representation in one invocation.
ninfer_softmax_attention_bf16_test	An Op, model or runtime test named for what it covers.
ninfer_softmax_attention_int8_test	An Op, model or runtime test named for what it covers.
ninfer_softmax_attention_fp8_test	An Op, model or runtime test named for what it covers.
ninfer_softmax_attention_nvfp4_test	An Op, model or runtime test named for what it covers.
ninfer_softmax_attention_k8v4_test	An Op, model or runtime test named for what it covers.
ninfer_sliding_window_attention_test	An Op, model or runtime test named for what it covers.
ninfer_kv_cache_append_test	An Op, model or runtime test named for what it covers.
ninfer_rmsnorm_rope_test	An Op, model or runtime test named for what it covers.
ninfer_context_kv_materialize_test_b8	KV materialisation against its FP64 oracle, swept by batch.
ninfer_context_kv_materialize_test_b1	KV materialisation against its FP64 oracle, swept by batch.
ninfer_context_kv_materialize_test_b2_3_4	KV materialisation against its FP64 oracle, swept by batch.
ninfer_context_kv_materialize_test_b5_6_7	KV materialisation against its FP64 oracle, swept by batch.
ninfer_kv_cache_append_nvfp4_test	An Op, model or runtime test named for what it covers.
ninfer_kv_cache_append_k8v4_test	An Op, model or runtime test named for what it covers.
ninfer_prepare_masked_block_test	An Op, model or runtime test named for what it covers.
ninfer_sparse_moe_test	An Op, model or runtime test named for what it covers.
ninfer_mtp_pack_test	An Op, model or runtime test named for what it covers.
ninfer_mtp_round_test	An Op, model or runtime test named for what it covers.
ninfer_speculative_round_test	An Op, model or runtime test named for what it covers.
ninfer_dynamic_grouped_conv_prepare_test	An Op, model or runtime test named for what it covers.
ninfer_linear_dynamic_grouped_conv_add_test	An Op, model or runtime test named for what it covers.
ninfer_linear_q4_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_q5_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_q6_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_q8_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_nvfp4_a16_test	An Op, model or runtime test named for what it covers.

ninfer_linear_nvfp4_a4_test	An Op, model or runtime test named for what it covers.
ninfer_linear_fp8_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_fp8_a8_test	An Op, model or runtime test named for what it covers.
ninfer_linear_bf16_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_split_test	An Op, model or runtime test named for what it covers.
ninfer_linear_split_multi_output_test	An Op, model or runtime test named for what it covers.
ninfer_linear_split_two_output_test	An Op, model or runtime test named for what it covers.
ninfer_qsa_tail_extract_test	An Op, model or runtime test named for what it covers.
ninfer_qsa_prepare_test	An Op, model or runtime test named for what it covers.
ninfer_qsa_indexer_test	An Op, model or runtime test named for what it covers.
ninfer_qsa_sparse_gqa_test	An Op, model or runtime test named for what it covers.
ninfer_linear_add_q4_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_add_q5_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_add_bf16_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_add_q8_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_add_nvfp4_test	An Op, model or runtime test named for what it covers.
ninfer_linear_add_fp8_test	An Op, model or runtime test named for what it covers.
ninfer_linear_pair_q8_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_swiglu_q4_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_swiglu_q8_a16_test	An Op, model or runtime test named for what it covers.
ninfer_linear_swiglu_nvfp4_test	An Op, model or runtime test named for what it covers.
ninfer_linear_swiglu_fp8_test	An Op, model or runtime test named for what it covers.
ninfer_gdn_input_proj_conv_snapshot_test	An Op, model or runtime test named for what it covers.
ninfer_gdn_input_proj_conv_record_test	An Op, model or runtime test named for what it covers.
ninfer_ple_conv_test	An Op, model or runtime test named for what it covers.
ninfer_ple_state_extract_test	An Op, model or runtime test named for what it covers.
ninfer_gated_delta_net_test	An Op, model or runtime test named for what it covers.
ninfer_gated_delta_net_replay_record_test	An Op, model or runtime test named for what it covers.
ninfer_gdn_replay_fold_test	An Op, model or runtime test named for what it covers.
ninfer_rmsnorm_test	An Op, model or runtime test named for what it covers.
ninfer_gated_rmsnorm_test	An Op, model or runtime test named for what it covers.
ninfer_gdn_control_test	An Op, model or runtime test named for what it covers.
ninfer_gdn_control_proj_test	An Op, model or runtime test named for what it covers.
ninfer_gdn_norm_control_proj_test	An Op, model or runtime test named for what it covers.
ninfer_hc_expand_norm_test	An Op, model or runtime test named for what it covers.
ninfer_hc_grouped_rmsnorm_test	An Op, model or runtime test named for what it covers.
ninfer_hc_combine_test	An Op, model or runtime test named for what it covers.

ninfer_hc_combine_norm_test	An Op, model or runtime test named for what it covers.
ninfer_hc_mix_test	An Op, model or runtime test named for what it covers.

The Python tests: 39 files

File	Trigger
tests/artifact/test_codecs.py	run by the hook only if listed; otherwise manual
tests/artifact/test_container.py	run by the hook only if listed; otherwise manual
tests/artifact/test_fp8_row.py	run by the hook only if listed; otherwise manual
tests/artifact/test_nf4.py	run by the hook only if listed; otherwise manual
tests/bench/test_request_log_reader.py	runs in the pre-commit hook
tests/bench/ttft/test_background.py	run by the hook only if listed; otherwise manual
tests/bench/ttft/test_campaign.py	run by the hook only if listed; otherwise manual
tests/bench/ttft/test_cases.py	run by the hook only if listed; otherwise manual
tests/bench/ttft/test_http_cancellation.py	run by the hook only if listed; otherwise manual
tests/bench/ttft/test_measurements.py	run by the hook only if listed; otherwise manual
tests/convert/quantization/test_fp8_row.py	run by the hook only if listed; otherwise manual
tests/convert/quantization/test_groupwise.py	run by the hook only if listed; otherwise manual
tests/convert/quantization/test_nvfp4.py	run by the hook only if listed; otherwise manual
tests/convert/quantization/test_stored_scale.py	runs in the pre-commit hook
tests/convert/test_cli.py	run by the hook only if listed; otherwise manual
tests/convert/test_official_recipes.py	run by the hook only if listed; otherwise manual
tests/convert/test_proposal.py	run by the hook only if listed; otherwise manual
tests/convert/test_qwen3_5.py	run by the hook only if listed; otherwise manual
tests/convert/test_qwen4_exp.py	run by the hook only if listed; otherwise manual
tests/convert/test_recipe.py	run by the hook only if listed; otherwise manual
tests/convert/test_resources.py	run by the hook only if listed; otherwise manual
tests/convert/test_sources.py	run by the hook only if listed; otherwise manual
tests/convert/test_verify_artifact.py	run by the hook only if listed; otherwise manual
tests/models/qwen3_5/test_tool_schema.py	run by the hook only if listed; otherwise manual
tests/release/test_activity_predicate.py	runs in the pre-commit hook
tests/release/test_calibration_corpus.py	runs in the pre-commit hook
tests/release/test_doc_links.py	runs in the pre-commit hook
tests/release/test_lane_figures.py	runs in the pre-commit hook
tests/release/test_launcher_generation.py	run by the hook only if listed; otherwise manual
tests/release/test_reference_audit.py	runs in the pre-commit hook
tests/release/test_retired_names.py	runs in the pre-commit hook
tests/release/test_subprocess_encoding.py	runs in the pre-commit hook

`tests/release/test_test_baseline_cache.py`	runs in the pre-commit hook
`tests/release/test_test_mutation.py`	runs in the pre-commit hook
`tests/release/test_text_encoding.py`	runs in the pre-commit hook
`tests/test_serve_corpus.py`	run by the hook only if listed; otherwise manual
`tests/text/test_chat_templates.py`	run by the hook only if listed; otherwise manual
`tests/text/test_json_schema.py`	run by the hook only if listed; otherwise manual
`tests/text/test_regex_choice.py`	run by the hook only if listed; otherwise manual

Probes, recipes and harnesses

None of these is a gate; each answers a question a gate cannot. The four newest exist because the measurement priority named the gaps they fill.

Tool	When it runs	What it answers
`tools/bench/report_latency_percentiles.py`	manual	p50/p90/p95/p99 for queue_wait, prepare, prefill, ttft, total, decode from a serve request log.
`tools/bench/probe_long_context_accuracy.py`	manual	A (length x depth) needle sweep with a fresh needle per cell; reports the largest length passing at every depth.
`tools/bench/probe_over_refusal.py`	manual	OR-Bench hard-safe and toxic rejection rates, reported as the pair.
`tools/bench/probe_concurrency_curve.py`	manual	Aggregate and per-request throughput against concurrency.
`tools/bench/lane_fingerprint.py`	manual (gate compares its record)	Every shipped lane on the cells the product depends on; records figures under the engine's hash.
`tools/bench/realtext_acceptance.py`	manual	Speculative acceptance on real text at a lane's own flags.
`tools/bench/probe_cancellation.py`	manual	Whether an abandoned request frees its slot.
`tools/bench/probe_cross_path_determinism.py`	manual	Cached against cold path on the same continuation.
`tools/bench/check_shared_prefix_reuse.py`	manual	Whether a prefix cache serves a shared prefix, by path rather than hit count.
`tools/bench/probe_prefill_floor.py`	manual	The prefill floor and the phase split, by name, from the request log.
`tools/bench/first_request_lane.py`	manual	The cold first request against later identical ones.
`tools/bench/warm_lane_sweep.py`	manual	`prepared` and TTFT against conversation size.
`tools/scripts/test_v3.cmd`	manual, and by the GPU workflow	Configures, builds and runs the whole C++ suite, then the baseline gate reads its log.
`tools/scripts/test_v3_asan.cmd`	manual	The host-only subset under MSVC AddressSanitizer, in its own tree.

`tools/scripts/test_v3_compute_sanitizer.cmd`	manual	A five-test device subset under compute-sanitizer memcheck.

The gate scripts

1. `documentation links`
2. `documentation citations`
3. `text encoding`
4. `FP8 band ladders`
5. `profile consistency`
6. `calibration corpus`
7. `stream-default ratchet`
8. `rule count`
9. `dead types`
10. `retired names`
11. `activity predicate`
12. `new duplication`
13. `port delta`
14. `skills`
15. `subprocess encoding`
16. `opencode plugin load`
17. `opencode claims-gate logic`